

Newnal

Intellectual Property Rights

*List of Technical Patents, Trademarks, Design Patents,
and Technical Whitepapers*

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1. Patent Application and Registration Status of Blockchain Labs, Newnal Inc.

Patent Category	Patent Office	Patent Applications	Patent Registrations
Technical Patents	USPTO (United States)	22	6
	WIPO / PCT (International)	5	5
	EPO (Europe)	1	1
	KIPO (Korea)	30	19
	(Total)	56	31
Trademarks	USPTO (United States)	3	2
	EPO (Europe)	1	0
	KIPO (Korea)	9	7
	WIPO (International)	1	1
	Other Countries	4	4
	(Total)	18	14
Design Patents	USPTO (United States)	3	2
	KIPO (Korea)	7	7
	CNIPA (China)	2	2
	(Total)	11	10
Total		85	55

2. Technical Patents

2.1 Core Patents

A) Core Blockchain Technology (InfraBlockchain)

BLOCKCHAIN SYSTEM TO WHICH PROOF-OF-TRANSACTION CONSENSUS ALGORITHM IS APPLIED, AND METHOD THEREFOR	
세계 최초로 개발된 가상화폐를 발행하지 않고 국가단위의 확장 가능한 스테이블코인 기반 블록체인 네트워크를 구축 가능하게 하는 Proof-of-Transaction(PoT) 블록체인 합의 알고리즘 기술 특허	
[USPTO] US-11636450-B2	
[WIPO/PCT] WO/2019/182202	
[KIPO] 10-225081	
MULTI BLOCKCHAIN NETWORK SYSTEM WITHOUT NATIVE CRYPTOCURRENCY BASED ON AGGREGATED PROOF-OF-TRANSACTION CONSENSUS	
가상화폐 발행없이 다수의 대규모 멀티 블록체인 네트워크를 운영할 수 있는 Aggregated Proof-of-Transaction (PoT) 멀티체인 합의 알고리즘 기술 특허. 병렬적으로 운영되는 다수의 블록체인 네트워크들로 구성된 초거대 연합형 멀티 블록체인 네트워크 구축을 가능하는 기술	
[USPTO] US-20240330915-A1	
[KIPO] 10-2023-0042651	

THE TRADING SYSTEM AND THE METHOD BASED ON A BLOCKCHAIN	
대규모 고빈도 거래소 구현을 위한 블록체인을 기반 안전하고 검증 가능한 온체인/오프체인 하이브리드 블록체인 토큰 거래 시스템 구축 기술 특허	
[WIPO/PCT] WO/2019/035573	
[KIPO] 10-2309819	

B) Blockchain-based Secure Communication & Messaging

ELECTRONIC DEVICE AND METHOD PROVIDING SECRET MESSENGER FUNCTION THROUGH END TO END ENCRYPTION BASED ON BLOCKCHAIN DID TECHNOLOGY	
블록체인 DID 기술에 기반한 종단간 암호화(E2EE)로 단말간 완전한 보안 통신을 제공하는 암호화 메시징 플랫폼 구축을 위한 기술 특허	
[USPTO] US-19/104055	
[WIPO/PCT] WO/2024/048838	
[KIPO] 10-2773087	

METHOD AND DEVICE FOR BUILDING DECENTRALIZED SECURE COMMUNICATION INFRASTRUCTURE NETWORK BASED ON BLOCKCHAIN AND DID TECHNOLOGY	
DID 기반 전화번호 생성 메커니즘을 이용한 확장성있는 블록체인 기반 탈중앙화 보안 통신 인프라 노드 네트워크를 구축을 위한 기술 특허	
[USPTO] US-19/059751	
[KIPO] 10-2025-0017836	

C) Personal Data Ownership & Web3 Data Infrastructure

BLOCKCHAIN-BASED WEB DATA OWNERSHIP AND DATA ACCESS MANAGEMENT METHOD AND SYSTEM FOR TRACKING AI DATA COLLECTION	
웹 데이터에 대한 검증 가능한 소유권 및 접근 권한을 관리하여, 개인이 AI의 개인 데이터 수집 및 활용 과정을 추적 통제할 수 있게 하는 블록체인 기반 시스템 구축을 위한 기술 특허	
[USPTO] US-20250055712-A1	
[EPO] EP-23195338.1	
[KIPO] 10-2023-0103842	

**DEVICE FOR PROVIDING BLOCKCHAIN DID-BASED MULTI CLOUD SERVICE
AND ITS OPERATING METHOD**

블록체인 및 DID 기반 멀티 클라우드 서비스를 제공하여, 개인 사용자의 탈중앙
신원(DID)으로 다수의 클라우드 환경에 개인데이터를 암호화된 형태로 분산 저장하고
개인데이터를 안전하게 관리할 수 있도록 하는 블록체인 기반 시스템 구축에 대한 기술
특허

[USPTO] US-20240414014-A1



[KIPO] 10-2023-0073929



**A DEVICE AND METHOD FOR OWNING AND TRADING VERIFIABLE PERSONAL
DATA THROUGH BLOCKCHAIN BASED PERSONAL WEB NODES**

블록체인 기반 개인 웹 노드를 통해 개인이 검증 가능한 개인 데이터를 저장하고 소유권을
증명할 수 있도록하여 데이터 주권을 사용자에게 부여하고, 개인 데이터를 안전하게 거래
가능하도록 하는 시스템 구축에 대한 기술 특허

[USPTO] US-19/038309



[KIPO] 10-2024-0012155



D) Personalized AI Model & AI Agent

PERSONALIZED AI ACTION GRAPH GENERATION MODEL BASED AI INFERENCE DEVICE	
개인 데이터로부터 학습된 개인화된 액션 그래프 모델을 생성하여 사용자별 의도와 행동을 예측하고 개인의 컨텍스트에 적합한 작업을 실행하는 AI 추론 시스템을 구현하는 기술 특허	
[USPTO] US-18/779899	
[KIPO] 10-2024-0090806	

DECENTRALIZED AI AGENT NETWORK SYSTEM BUILT UPON BLOCKCHAIN-BASED WEB3 AI-ID AND WEB3 AI-CONTRACT FOR VERIFIABLE CONFIDENTIAL AI AGENT COMPUTING	
블록체인 기반 Web3 AI-ID 및 AI-Contract 위에 구축되는 탈중앙화 AI 에이전트 네트워크 구현 기술로, AI 에이전트 간의 검증 가능하고 기밀성 있는 컴퓨팅을 지원하는 기술 특허	
[USPTO] US-19/303959 US-63/764594	
[KIPO] 10-2025-0107438	

BLOCKCHAIN-BASED AI AGENT SERVICE MARKETPLACE SYSTEM INVOLVING PERSONAL AI AGENTS AND SERVICE AI AGENTS UTILIZING PERSONAL DATA

개인 AI 에이전트와 서비스 AI 에이전트가 오픈 마켓을 통해 개인데이터를 거래하고 개인 데이터를 활용한 서비스를 거래하는 블록체인 기반 마켓플레이스 플랫폼 구축을 위한 기술 특허

[USPTO] US-19/702994
US-63/865619



[KIPO] 10-2026-0079294



A DEVICE AND METHOD FOR EXTRACTING AND STRUCTURING VERIFIABLE PERSONAL DATA FROM USER INPUT AND OUTPUT DATA CAPTURED ON A DEVICE BASED ON A MULTI-MODAL AND LANGUAGE AI MODEL

멀티 모달 LLM AI 모델을 이용하여 사용자의 기기 내 입출력 데이터로부터 검증 가능한 개인 데이터를 추출하고 구조화하는 시스템을 구현하는 기술 특허

[USPTO] US-19/041098



[KIPO] 10-2024-0014525



E) Personal AI Operating System & Devices

MOBILE DEVICE FOR OPERATING PERSONALIZED ON-DEVICE AI AGENT BASED ON PERSONAL DATA	
사용자의 개인 데이터를 기반으로 개인화된 AI 에이전트를 단말에서 직접 구동해, 추론과 개인화를 온디바이스로 처리함으로써 프라이버시와 데이터 통제권을 보장하는 모바일 디바이스 기술 특허	
[USPTO] US-19/059568	
[KIPO] 10-2025-0019132	
USER INTERFACE FOR RUNNING A PERSONALIZED AI-DEDICATED OPERATING SYSTEM ON VARIOUS TYPES OF USER DEVICES	
개인화된 AI 전용 운영체제를 다양한 종류의 사용자 디바이스에서 일관되게 구동할 수 있도록 하는 AI 전용 기기를 위한 사용자 인터페이스 특허	
[USPTO] US-19/277871	
[KIPO] 10-2025-0098140	
BLOCKCHAIN-BASED AI COMPUTING MOBILE OPERATING SYSTEM FOR SECURE DECENTRALIZED PERSONAL DATA AND PERSONALIZED AI AGENT SERVICES	
개인 데이터를 안전하고 탈중앙적으로 처리하고 개인화된 AI 에이전트 서비스를 단말에서 제공하는 블록체인 기반 모바일 운영체제 기술 특허	
[USPTO] US-63/884059	

F) Decentralized Identity (DID) & Verifiable Credentials

DID AUTHENTICATION METHOD USING SMART CARD AND SMART CARD DEVICE	
물리적 보안 IC 칩를 통해 탈중앙 신원을 생성하고 인증을 수행하는 블록체인 기반 DID 인증 시스템 및 스마트 카드 장치에 대한 기술 특허	
[USPTO] US-18/709142	
[WIPO/PCT] WO/2023/085802	

METHOD AND SYSTEM FOR PROVIDING CERTIFICATION OF VACCINE INOCULATION AND POST-INOCULATION MANAGEMENT	
블록체인 기반 검증 가능한 자격증명을 활용해 백신 접종 증명서를 발급·검증하고 접종 후 이력을 관리하는 시스템 구축 기술 특허	
[USPTO] US-20240185994-A1	
[WIPO/PCT] WO/2022/196851	
[KIPO] 10-2488139	

IDENTITY MANAGEMENT SYSTEMS AND METHODS USING DID BASED ON BLOCKCHAIN TECHNOLOGY	
블록체인 기반 탈중앙 식별자(DID)를 이용해 사용자가 자신의 신원을 자기주권적으로 통제할 수 있게 하는 신원 관리 시스템 및 방법	
[KIPO] 10-2022-0100263	

2.2 Periphery Patents

[USPTO] US-63/763955

A METHOD AND SYSTEM FOR COLLECTING AND INTEGRATING PAST, PRESENT AND FUTURE PERSONAL DATA TO IMPLEMENT PERSONALIZED AI MODEL AND AI AGENT ENGINE



[KIPO] 10-2024-0014525

A DEVICE AND METHOD FOR EXTRACTING AND STRUCTURING VERIFIABLE PERSONAL DATA FROM USER INPUT AND OUTPUT DATA CAPTURED ON A DEVICE BASED ON A MULTI-MODAL AND LANGUAGE AI MODEL



[USPTO] US-19/303959

METHOD AND DEVICE FOR CONFIGURING AND OPERATING SELF-EVOLVING PERSONAL AI AGENT BASED ON PERSONAL DATA



[KIPO] 10-2026-0012833

METHOD AND DEVICE FOR CONFIGURING AND OPERATING SELF-EVOLVING PERSONAL AI AGENT BASED ON PERSONAL DATA



[USPTO] US-19 (application in progress)

BLOCKCHAIN-BASED HOME NODE DEVICE AND SERVER-BASED OPERATING SYSTEM FOR PROVIDING PERSONAL DATA SERVICES AND PERSONAL AI AGENT SERVICES



[KIPO] 10-2026-0013613

BLOCKCHAIN-BASED HOME NODE DEVICE AND SERVER-BASED OPERATING SYSTEM FOR PROVIDING PERSONAL DATA SERVICES AND PERSONAL AI AGENT SERVICES



[USPTO] US-63/763302

METHOD AND SYSTEM FOR ADAPTIVE FACIAL REGION BLENDING IN VIDEO SYNTHESIS



[KIPO] 10-2249864

SERVER AND METHOD FOR CREDIT TRANSACTION USING BLOCKCHAIN NETWORK



[KIPO] 10-2023-0165942

HIGH-PERFORMANCE CONSENSUS ALGORITHM SIMULATOR



[KIPO] 10-2700653

THE SYSTEM AND THE METHOD FOR MANAGING DISTRIBUTION HISTORY USING NON-FUNGIBLE TOKEN



[KIPO] 10-2022-0130057

SYSTEM AND METHOD FOR TRADING BASED ON BLOCKCHAIN TECHNOLOGIES



[KIPO] 10-2613633

AI MODEL TRAINING APPARATUS AND METHOD FOR READING TEST RESULTS OF A DIAGNOSTIC KIT



[KIPO] 10-2478963

A SYSTEM AND METHOD FOR ISSUING AND VERIFYING DIGITAL VACCINATION CERTIFICATES



[KIPO] 10-2490640

ELECTRONIC DEVICE AND METHOD FOR PROVING PRIVATE INFORMATION BETWEEN INDIVIDUALS BASED ON BLOCKCHAIN TECHNOLOGY



[KIPO] 10-2591773

METHOD AND SYSTEM FOR CERTIFICATING COVID-19 ANTIBODY



[KIPO] 10-2022-0100264

AI-BASED IDENTITY RECOGNITION METHOD AND ELECTRONIC DEVICE THEREOF



[KIPO] 10-2022-0108933

BLOCKCHAIN DID-BASED REAL-TIME REAL ESTATE REGISTRATION MANAGEMENT SYSTEM



[KIPO] 10-2023-0051359

VEHICLE REMOTE CONTROL AND VEHICLE INFORMATION MANAGEMENT METHOD USING VEHICLE-DID BASED ON BLOCKCHAIN NETWORKS AND THE VEHICLE DEVICE THEREOF



3. Trademarks

Title	Class	Application Number
Newnal	09, 35, 36, 38, 42	[US] US 98/180961 [EU] EP 18943637 [CH,GB] WIPO 1773137 [KR] 40-2023-0167191
Metacircle	9	[US] US 99/290317 [EU] EP 19219923 [KR] 40-2025-0125955
COOV	09,42	[KR] 40-2021-0065186 [KR] 40-2021-0065182
InfraBlockchain	42	[KR] 40-2023-0122739
BlockChat	09, 38, 42	[KR] 40-2022-0139282 [KR] 40-2022-0139284 [KR] 40-2022-0207516 [GB,FR,ES,DE,IT,NL,BR,AU,AT,TR,RU,PT,ID,IN,SA,PH] WIPO 1710775 [HK] 306125436 [AR] 4196441 [TW] 111090047
BLOCKCHAIN CHIP Powered by Blockchain Labs	9	[KR] 40-2023-0100452

4. Design Patents

[KIPO] 30-2024-0004156

Devices with a rollable display



[KIPO] 30-2024-0004155

Devices with a rollable display – unrolled



[KIPO] 30-2024-0004155

Devices with a rollable display – rolled



[KIPO] 30-2025-0015837

Mobile phone device (Newnal Phone V1)



[KIPO] 30-2025-0015838

Mobile phone device – AI jog button (Newnal Phone V1)



[USPTO] US-30/015605

Mobile phone device (Newnal Phone V1)



[CNIPA] 202530452354.9

Mobile phone device (Newnal Phone V1)



[CNIPA] 202530510789.4

Mobile phone device (Newnal Phone Light)



[KIPO] 30-2025-0029489

Mobile phone device (Newnal Phone Light)



[USPTO] US-30/018308

Mobile phone device (Newnal Phone Light)



[KIPO] 30-2025-28631

Newnal Metacircle UI design – Homescreen, Lockscreen



[USPTO] US-30/015673

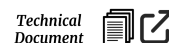
Newnal Metacircle UI design – Homescreen, Lockscreen



5. Technical Whitepapers

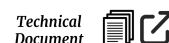
Newnal Web3 ai Technical Whitepaper

'My own ai' (another-i) model that learns from personal data



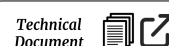
Newnal Web3 ai OS Technical Whitepaper

Blockchain-based 'Appless OS' architecture for personal AI agents having Web3 ai-ID and Web3 ai-Contract



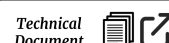
Newnal Web3 Telecom Technical Whitepaper

Next-generation decentralized and secure communication protocol including AI-involved calls



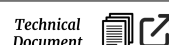
Newnal ai Agent Place Technical Whitepaper

Marketplace for personal data and ai(another-i) agent service exchange



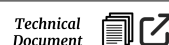
Newnal InfraBlockchain Technical Whitepaper

Stablecoin-native public federated multi-blockchain network and verifiable web data infrastructure



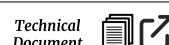
Universal Resource Auth (UR-Auth) Protocol and System Architecture Technical Whitepaper

Blockchain-based Web-scale Data Ownership Management Technology



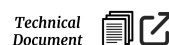
InfraBlockchain Technical Whitepaper

Stablecoin as system tx fee token, Transaction-as-a-Vote, Proof-of-Transaction(PoT) consensus algorithm

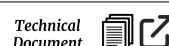


ON/OFF-Chain Hybrid Token Exchange System Technical Whitepaper

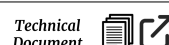
Blockchain-based large-scale high frequency trading system technology



InfraBlockchain-Substrate Multi-Blockchain Developer Documentation



InfraBlockchain-Antelope Developer Documentation



InfraDID Method Specification***InfraDID W3C DID Spec. Registry (did:infra)******InfraBlockchain Open-Source Technology***